



MVP Build Specification v3

Architecture, functionality and business rules — written for AI coding agents

MARKET	Liberia only (signup geo-restricted)
MODES	Relationship (free) · Casual Connection (paid)
MOBILE MONEY	Orange Money · MTN — Day / 7-Day / Monthly, admin-verified
CARDS	Visa / Mastercard — recurring, Weekly minimum
BACKEND	Supabase + Next.js
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DATE	5 October 2026

How to read this document

This is the build specification for WeKonnctz, an adults-only connection platform for Liberia. It is written for the owner and for AI coding agents. It defines the architecture, every feature, the business rules, and the things an agent must never guess.

Marker	Meaning
[NEW]	New or rewritten in v3.
[CHANGED]	Changed from the original product spec, for a reason stated in place.
[DECISION]	A product or business decision the owner has not made. An AI agent must stop and ask — never invent a value.
[VERIFY]	A factual assumption (provider capability, SMS delivery, legal requirement) to confirm before the sprint that depends on it.

Decisions confirmed by the owner

Topic	Decision
Product	Two modes on one account: Relationship (free) and Casual Connection (paid).
Market	Liberia only. No diaspora. Location checked at signup only.
Mobile money	Orange Money and MTN Mobile Money. Day, 7-Day and Monthly passes, one-off. Member pays outside the app, submits a payment claim in the app, an admin verifies it manually, and access activates directly.
Cards	Visa / Mastercard recurring subscriptions. Minimum plan: Weekly. No manual verification. Processor not yet chosen.
Removed	Coins / wallet, voucher codes, any WhatsApp channel.
Backend	Supabase (Auth, Postgres with RLS, Storage, Realtime, Edge Functions, pg_cron) with Next.js.
Verification	Strong verification is a product pillar.

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1 What changed in v3

Payments were redesigned around what is available in Liberia today. Everything else carries over from v2, including all audit fixes, which are built into the sections below.

Area	v2	v3
Mobile money	Direct API integration through a provider adapter	Manual: member submits an in-app payment claim; an admin verifies it against the merchant wallet. APIs plug in later behind the same adapter. [NEW]
Cards	One-off passes, same as mobile money	Recurring subscriptions, Weekly minimum, hosted checkout and webhooks. [NEW]
Subscription states	Renewal states reserved for later	CANCELLED and PAYMENT_FAILED are live for card renewals.
Coins, vouchers	Under discussion	Removed. Not to be built.
WhatsApp	Not specified	Excluded: Meta's WhatsApp Business policy prohibits dating services (§24).
Architecture	Implicit	New §4: components, module boundaries, both payment flows.
Data model	—	New: payment_claims, merchant_accounts, card_customers. Subscriptions gain source and renewal fields.
Admin	Payments list	New payment claims queue with a verification checklist.
Scope control	—	New §24: what agents must not build.

2 Product definition

WeKconnectz is a premium, adults-only (18+) connection platform for people in Liberia, with two distinct modes sharing one account, one profile and one verification.

	Relationship	Casual Connection
Price	Free	Mobile money pass (Day / 7-Day / Monthly) or card subscription (Weekly / Monthly)
Discovery	Swipe-style Discover feed	“Available Now” pool
Contact	Mutual like → match → chat	Message request → accept → chat
Eligibility	Active, verified, 3 approved photos	Same + active pass + Casual intent + availability set
Purpose	Intentional dating	Adults who want to meet socially with less ceremony

Core principle (kept, strengthened)

A subscription grants access to a private member pool and messaging tools. It does not purchase a person, a meeting, a sexual service, a response, or any outcome. Any member offering or requesting sex for money, gifts or other compensation is removed. **[CHANGED]**

Prohibited use **[NEW]**

- Offering, advertising or requesting sex in exchange for money, airtime, mobile money, gifts, transport fare or any other compensation.

- Prices, rates, “services”, phone numbers, WhatsApp handles or social links in bios, photos or message requests.
- Anyone under 18, or content depicting anyone under 18.
- Nudity or sexually explicit photos.
- Impersonation, fake profiles, using another person’s photos.
- Requesting money from other members for any reason (romance-scam pattern).

3 Market access: Liberia-only

[NEW] Owner decision: block at signup only. Once registered, a user keeps access when travelling.

Rules

1. **Signup IP check.** Account creation is allowed only when the request’s country, read from the hosting edge header (x-vercel-ip-country on Vercel or CF-IPCountry behind Cloudflare), equals LR.
2. **Liberian phone required.** The OTP phone number must be a +231 number. This is the control that actually enforces Liberia-only, because OTP needs a working Liberian SIM.
3. **Order.** Run the IP check before sending any OTP. That stops SMS spend on blocked traffic.
4. **Server-side only.** The check runs inside the registration server action. Middleware may pre-screen, but the authoritative check is never client-side.
5. **Record.** Store each attempt in geo_checks: ip_country, phone_country, result, timestamp. Store the country code, not the raw IP, unless OD-12 says otherwise.
6. **No re-checks after signup** (owner decision).
7. **Feature flag.** GEO_ENFORCEMENT_MODE = SIGNUP_ONLY | EVERY_SESSION in app_settings, so stricter enforcement can be switched on later without a code rewrite.
8. **Blocked screen copy:** “This platform is currently available only in Liberia.” No detail on how the check works.
9. **Public pages stay global.** Landing, About, Safety, Terms and Privacy remain reachable from anywhere. Only registration is restricted.

Audit note. IP checks at signup are bypassed by any VPN. On its own, signup-only IP blocking stops casual foreign signups and little else. The +231 OTP requirement is what does the real work — keep it non-negotiable. Consequence of your no-diaspora decision: a Liberian abroad on a roaming +231 SIM passes the phone check but fails the IP check, and is blocked. That is consistent with the decision; confirm it is intended.

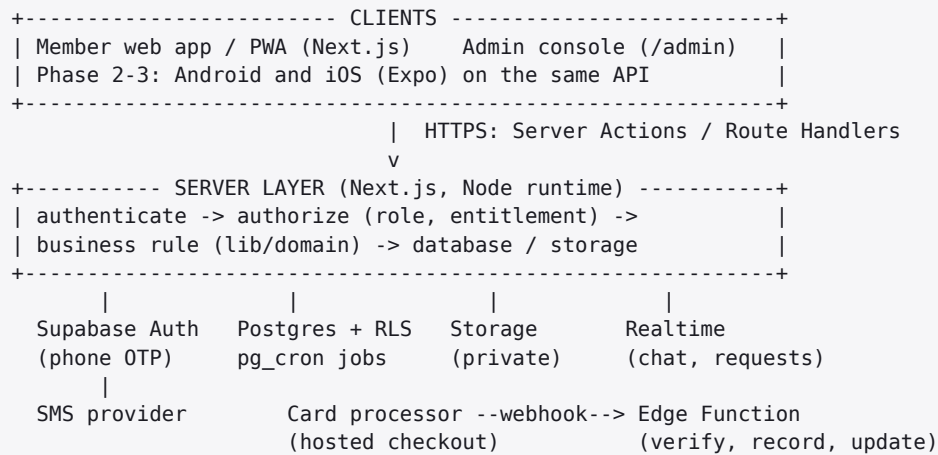
Verify before Sprint 1 **[VERIFY]**

- SMS OTP delivery to Orange and Lonestar Cell MTN numbers through the SMS provider wired into Supabase Auth: deliverability, latency and per-message cost.
- Whether both edge headers are present on the chosen hosting plan.

4 System architecture

[NEW] One Next.js application serves members and staff. All business logic runs on the server. Supabase provides auth, data, files, realtime and scheduled jobs. Card payments arrive by processor webhook; mobile money arrives by admin approval.

Components



Module boundaries

Path	Responsibility	Rule
lib/domain/	Pure business rules: eligibility, entitlement, stacking, state transitions	No Next.js or Supabase imports, so mobile can reuse it
lib/payments/claims/	Mobile money claim submission and validation	Activation only via <code>approve_payment_claim()</code>
lib/payments/card/	CardProcessor adapter, checkout, cancellation	Processor-specific code stays inside its adapter
lib/auth/, lib/permissions/	Session, roles, staff MFA checks	Called at the top of every server action
lib/storage/	Upload pipeline, signed URLs	Never returns a storage path to a client
app/(public), app/(member), app/admin	Routes and UI	No business logic in components
supabase/migrations/	Schema, RLS policies, database functions	The only way the schema changes
supabase/functions/	Card webhooks, scheduled tasks	Signature check first, idempotent

Mobile money flow (manual verification)

MEMBER	APP / SERVER	ADMIN
choose pass + provider	-> show merchant number, exact amount, reference code [VERIFY]	
pay via USSD / wallet app (outside WeKconnectz)		
submit transaction ID	-> validate: amount = plan price, txn ID unused, pending limit	
+ screenshot	store claim PENDING_REVIEW	----> claims queue
		find txn ID + amount in merchant wallet records
	approve_payment_claim() <-----	approve reject needs info
	one transaction: claim APPROVED,	
	payment SUCCEEDED, access granted,	
	audit log, notification	
notified, access live <-		

Card flow (recurring)

MEMBER	APP / SERVER	CARD PROCESSOR
choose Weekly/Monthly	-> createCheckout()	-----> hosted checkout page
enter card on the processor's page (no card data on WeKconnectz servers)		
	webhook Edge Function <-----	subscription / payment events
	verify signature, log event,	
	update subscription	
renewal:	period extended <-----	renewal charge succeeded
failure:	PAYMENT_FAILED -> grace -> EXPIRED	
cancel:	CANCELLED, access until period end	

Platform phases

Phase	Client	Notes
1	Responsive web app / PWA	Ships the full MVP including admin.
2	Android (Expo / React Native)	After the web product is validated. Check Google Play dating policies first [VERIFY]
3	iOS (Expo / React Native)	Check App Store dating-app guidelines first [VERIFY]

5 Tech stack for AI agents

Matches the stack already in use (Next.js, TypeScript, Tailwind, Supabase, Cloudflare). Each row carries a rule the agent must follow. Pin the latest stable version of each package at project start and record versions in `CLAUDE.md`.

Layer	Choice	Rule for the agent
Language	TypeScript, <code>strict: true</code>	No any without a justifying comment. No <code>@ts-ignore</code> .
Framework	Next.js App Router	Server Components by default. Mutations only through Server Actions or Route Handlers.
UI	Tailwind CSS + shadcn/ui	No second component library. Shared primitives in <code>components/ui</code> .
Forms	React Hook Form + Zod	One Zod schema per form, reused on the server. Server validation is authoritative.
Client data	TanStack Query	Only for client-interactive lists (discovery, messages, admin tables).
Motion	Framer Motion	Sparingly; respect <code>prefers-reduced-motion</code> .
Database	Supabase Postgres	Schema changes only as SQL migrations via Supabase CLI. Never edit schema in the dashboard. Regenerate types after each migration.
Auth	Supabase Auth, phone OTP	+231 numbers only. Email optional, for receipts. [VERIFY] SMS provider coverage.
Authorization	Postgres RLS + server checks	RLS enabled on every table at creation, with explicit policies. Entitlements checked server-side.
Storage	Supabase Storage, private buckets	Buckets: photos-quarantine, photos, verification, payment-evidence. No public bucket. Images served as short-lived signed URLs (default 120 s).
Image processing	sharp (Node runtime)	Validate type by magic bytes, cap size, resize, convert to WebP, strip all EXIF metadata.
Realtime	Supabase Realtime	Messages and request notifications. Channels authorised by conversation membership.
Scheduled jobs	pg_cron	Expire availability windows, passes and pending payments. Access checks also compare timestamps at query time — cron is cleanup, not the gate.
Server functions	Supabase Edge Functions	Card processor webhooks: verify signature, idempotent on processor event ID.
Payments	Two paths in <code>lib/payments</code>	Mobile money: in-app claims approved by an admin (\$16). Cards: CardProcessor adapter, hosted checkout, webhooks; processor not chosen.
Geo	Edge country header	Read only in the registration server action (\$3).
Rate limiting	Postgres counters (Upstash Redis optional)	OTP sends, message requests, likes, reports, uploads.
Errors	Sentry	Scrub phone numbers, DOB, storage paths, message text.
Analytics	PostHog (optional)	No message content or sensitive profile fields.
Testing	Vitest · Playwright · pgTAP	Unit tests for every business rule; e2e for each sprint's flow; pgTAP tests for every RLS policy.
Hosting	Vercel + Supabase Cloud	Alternative: your Hetzner VPS. [DECISION] OD-11.
Mobile (later)	Expo / React Native	Reuses domain logic, schemas, API.

6 Agent operating rules & repo docs

An AI agent given “build this dating app” produces a tangle. These files and rules exist before significant code is written.

Documents in the repository

File	Contents
CLAUDE.md	Stack and versions, commands (dev, test, migrate, typegen), folder conventions, the hard rules below, current sprint pointer.
docs/BUSINESS_RULES.md	Section 19 verbatim. Numbered rules referenced in code comments and tests (e.g. // BR-26).
docs/DATA_MODEL.md	Section 18, kept in sync with migrations.
docs/SECURITY.md	Section 22 and the request-flow rule.
docs/DECISIONS.md	Section 25 open decisions; resolved ones recorded with date and reason.
docs/sprints/SPRINT-XX.md	Scope, out-of-scope, acceptance criteria checklist for one sprint.

Hard rules for every agent session

1. Work on one sprint at a time. At the end, run tests, tick the acceptance checklist and stop for owner confirmation before starting the next.
2. Never create a table without enabling RLS and writing its policies in the same migration.
3. Never use the Supabase service-role key outside modules marked `import "server-only"`.
4. Never decide entitlement (premium, availability, verification, role) from client state. Read it from the database on the server.
5. Never return a storage path to the client. Return a signed URL only after an access check.
6. Every sensitive admin mutation writes `audit_logs` in the same transaction.
7. Never log phone numbers, dates of birth, selfie paths or message text.
8. No new dependency without stating why in the PR/commit message.
9. If a value is marked [DECISION] or [VERIFY], stop and ask. Do not insert a placeholder number into business logic; use `app_settings` keys with no default and fail loudly.
10. Each business rule touched gets a test that names its rule number.
11. Casual access from mobile money is granted only by the database function `approve_payment_claim()`, called from an admin action. No other code path may create it.
12. Never build anything listed in §24 (WhatsApp, coins, vouchers, automatic claim approval).

The request-flow rule (kept)

```
CLIENT -> SERVER ACTION / ROUTE -> AUTHENTICATE -> AUTHORIZE -> BUSINESS RULE -> DB / STORAGE

Never: CLIENT -> "if (isPremium) showPhoto()" <- this is not security
```

7 Roles & permissions

Capability	USER	MODERATOR	ADMIN	SUPER_ADMIN
Use the member app	✓	Separate account	Separate account	Separate account
Review photos		✓	✓	✓
Review verification selfies (each view logged)		✓	✓	✓
Handle reports, add notes		✓	✓	✓
Suspend users (time-limited)		✓	✓	✓
Ban / restore users			✓	✓
View subscriptions & payments			✓	✓
Approve / reject mobile money payment claims			✓	✓
Modify subscriptions, issue refunds			✓	✓
Analytics & settings			✓	✓
Manage staff, change roles				✓
View audit logs			✓ (read)	✓
Feature flags & system config				✓

[NEW] Staff accounts are separate from personal member accounts and require MFA (Supabase TOTP). A staff member never moderates a report involving their own member account.

8 Account states

State	Meaning	Entered from	Can do
PENDING	Registered, onboarding incomplete or under review	Signup	Finish onboarding; edit own profile
ACTIVE	Onboarding done, verification approved	PENDING, SUSPENDED	Everything their entitlements allow
SUSPENDED	Temporary restriction with <code>suspended_until</code>	ACTIVE	Log in, read messages, view own account; no discovery, no sending
BANNED	Permanent	ACTIVE, SUSPENDED	Cannot authenticate. Phone hash added to blocklist.
DELETED	User requested deletion	Any except BANNED	Hidden immediately; data purged after retention period [DECISION] OD-7

[CHANGED] Suspensions auto-lift when `suspended_until` passes. A banned phone number cannot register again.

9 Age & identity verification

[CHANGED] Verification is required for everyone — Relationship and Casual — consistent with strong verification as a product pillar.

Layers

Layer	What happens	When
1. Date of birth	Entered once at the age gate. Under 18 → blocked. Locked after submission; only an admin can correct it, with an audit entry.	Onboarding step 2
2. Liberian phone OTP	One account per verified phone. Ties the account to a real SIM.	Onboarding step 3
3. Liveness selfie	App shows a randomly chosen pose instruction (e.g. a hand gesture). User takes the selfie in-app (camera capture, not gallery upload).	Onboarding step 10
4. Human review	Reviewer checks: pose matches the instruction; face matches the profile photos; person plainly appears adult. Doubt about age → reject and escalate.	Verification queue
5. ID document	Recommended before Casual eligibility. [DECISION] OD-5	Before first Casual availability

Verification statuses (kept)

NOT_STARTED → PENDING → VERIFIED | REJECTED. A rejected user may resubmit; repeated rejections (count set in app_settings) escalate to an admin.

Selfie handling **[NEW]**

- Stored in the private verification bucket. Never shown to members, never used as a profile photo.
- Only MODERATOR and above can view it, through a signed URL; each view writes an audit entry.
- Retention after a decision: **[DECISION]** OD-6. Recommendation: delete the image 90 days after approval, keep the decision record.

10 Registration & onboarding

#	Screen	Contents	Change
1	Welcome	Logo · “Meet with intention.” · Create account · Log in	
2	Age gate	“You must be 18 or older.” Date of birth (only place it is asked).	[CHANGED]
3	Phone	+231 number → server geo check → OTP → verify.	[CHANGED]
4	Community rules & terms	Plain-language rules (no paid sex, no minors, no contact info, no money requests), Terms, Privacy. Acceptance stored with document version.	[NEW]
5	Basic profile	Display name · gender · interested in (one or more genders) · area (county + community from a list).	[CHANGED]
6	Intent	Relationship · Casual Connection · Both. Casual requires a pass later; choosing it here only sets intent.	
7	Interests	At least 3 from an admin-managed list.	

#	Screen	Contents	Change
8	Bio	Up to 500 characters. Server rejects phone numbers, links, handles and price patterns.	[CHANGED]
9	Photos	3 required (max 6 recommended). Primary must show the face clearly.	
10	Verification selfie	In-camera capture with a random pose prompt.	[CHANGED]
11	Under review	"We're reviewing your profile." Shows progress of photos and verification. User can edit while waiting.	[NEW]

Progress is saved per step; a user who leaves returns to the first incomplete step. Account becomes ACTIVE when verification is VERIFIED and at least 3 photos are APPROVED.

11 Photos & privacy

Upload pipeline [NEW]

```
client -> request signed upload URL (server checks count & rate limit)
        -> upload to photos-quarantine/
        -> server job: magic-byte check, size cap, resize, WebP, strip EXIF
        -> move to photos/ , status PENDING_REVIEW
        -> moderator APPROVED | REJECTED (reason required)
```

Photo states (kept)

UPLOADING · PENDING_REVIEW · APPROVED · REJECTED · HIDDEN · DELETED. Only APPROVED photos count toward the minimum of 3. If a user drops below 3 approved photos, they leave discovery and their availability is set to UNAVAILABLE automatically.

Photo content rules

- Primary photo shows the user's face clearly; no group photo as primary.
- No nudity, underwear-focused or sexually suggestive poses.
- No text overlays, phone numbers, prices or social handles.
- No children in any photo.
- No photos of other people presented as the user.

Who sees what [CHANGED]

Viewer	Member profile data	Photos	Availability
Logged-out visitor	Nothing (no public profiles)	None	None
Member not yet ACTIVE	Nothing	None	None
ACTIVE member, Relationship	Profiles in Relationship discover / matches	Per OD-3	Never
ACTIVE member with Casual pass	Profiles in the Available pool	Approved photos of pool members	Yes, inside the pool
Blocked pair (either direction)	Nothing	None	None

OD-3 — photo visibility in free Relationship mode. The original spec hides photos from non-paying users, but Relationship is now free. Options: (A) Relationship members see each other's approved photos; the Casual pass sells access to the pool and messaging, not photos. (B) Relationship shows the primary photo only; the full gallery needs a pass. (C) Separate photo sets per mode. Recommendation: A — simplest, and a free dating mode without photos will not retain users. Your call.

12 Availability

States (kept)

State	Meaning
UNAVAILABLE	Not in the pool.
AVAILABLE	In the pool for the current window.
PAUSED	Hidden, schedule kept.

Eligibility to become AVAILABLE

- Account ACTIVE and not suspended.
- Verification VERIFIED (and ID check, if OD-5 adopts it).
- 3 approved photos.
- Intent includes Casual.
- Active Casual pass (kept from original — both sides of the pool pay).

Windows [CHANGED]

- “Available now” opens a window from now to a chosen end time.
- “Schedule” sets a future start and end.
- One active-or-scheduled window per user.
- Maximum window length and how far ahead a window can be scheduled: **[DECISION]** OD-8 (recommendation: 12 hours max, 7 days ahead).
- A user is in the pool only when: status = AVAILABLE **and** now is between start_at and end_at **and** the pass is active. This is computed at query time; the cron job only tidies status afterwards.
- If the pass expires mid-window, the user leaves the pool immediately.

13 Casual discovery

Page: **Available Now**. Requires an active pass.

Element	Spec
Filters	Area (county/community) · age range · gender (pre-set from “interested in”) · interests · availability window. [CHANGED] “Verified” filter removed: everyone in the pool is verified.
Card	Primary photo · name, age · area · “Available until 11 PM” · verified badge · View profile
Exclusions	Self · blocked pairs · users not matching each other’s “interested in” · suspended or under-review accounts · users whose message setting is Nobody (hidden, not shown greyed out)
Ordering	Rotate fairly (e.g. recently available first, then randomised within buckets) so the same few profiles do not dominate. Exact algorithm is an implementation detail.
Pagination	Cursor-based, 20 per page.
Saved profiles	Bookmark a member; saved list shows only those still eligible to be seen.

14 Messaging

Casual: message requests

1. Sender must have an active pass and be VERIFIED. Recipient must be AVAILABLE when the request is sent.
2. First contact is a request: text only, up to 300 characters, no links, numbers, handles or price patterns (server-validated).
3. Recipient: **Accept** (conversation opens) · **Decline** · **Block**.
4. One pending request per sender → recipient. After a decline, the sender cannot request the same person again for a cool-down period **[DECISION]** OD-9.
5. Daily request cap per sender **[DECISION]** OD-9.
6. Recipient settings **[CHANGED]**: **Anyone in the pool** or **Nobody**, plus individual blocks. (“Verified only” removed — redundant.)

Relationship

Mutual like creates a match; the match opens a RELATIONSHIP conversation. Free. No messaging before a match.

Rules common to both

- MVP messages are text only. No images, voice notes or links. **[NEW]**
- Conversations are strictly one-to-one.
- Messages pass the same detection used for bios (§17). Matches are flagged for review, not auto-blocked.
- Blocking or unmatching closes the conversation for both sides.
- Pass expiry: CASUAL conversations become read-only for any participant without an active pass. History is kept. **[CHANGED]**
- Read receipts via read_at; typing indicators optional (Realtime presence).

15 Relationship mode

- Discover feed: one profile at a time, Like or Pass. Filters: area, age range, gender (from “interested in”), interests.
- Eligibility to appear: ACTIVE, VERIFIED, 3 approved photos, intent includes Relationship. **[CHANGED]** (verification now required here too).
- Likes are unique per (sender, receiver). A mutual like creates a match inside one database function, atomically, so two simultaneous likes cannot produce two matches.
- Match rows store the pair once, ordered (`user_a_id < user_b_id`).
- Statuses: ACTIVE · UNMATCHED. Unmatching hides the conversation from both.
- Passed profiles return to the feed after a cool-down (value in `app_settings`).
- Daily like cap for free users: **[DECISION]** OD-10. Useful against spam bots; also a future upsell lever.
- Switching modes never bypasses the Casual pass (kept).

16 Subscriptions & payments

[NEW] Two ways to pay for Casual access. Mobile money is manual until Orange Money and Lonestar Cell MTN provide APIs. Cards are automatic and recurring.

Terminology. “Casual access” = an active entitlement from either source: a mobile money pass or a card subscription. Wherever this spec says “pass”, it means Casual access from either source.

Plans

Source	Plan	Length	Renews	Price
Mobile money	Day Pass	24 hours	No	[DECISION] OD-1
Mobile money	7-Day Pass	7 days	No	[DECISION] OD-1
Mobile money	Monthly	30 days	No	[DECISION] OD-1
Card	Weekly	7 days	Yes	[DECISION] OD-1
Card	Monthly	30 days	Yes	[DECISION] OD-1, OD-15

Plans and prices live in `subscription_plans`, editable by admins. Currency: **[DECISION]** OD-2.

Mobile money: payment claims

1. Member picks a mobile money plan and a provider (Orange Money or MTN Mobile Money).
2. App shows: the WeKconnectz merchant number or code for that provider (from `merchant_accounts`), the exact amount, and a short reference code (e.g. WK-7F3K) to enter in the payment note if the provider allows it **[VERIFY]** OD-22.
3. Member pays outside the app (USSD or wallet app).
4. Member submits a claim: transaction ID (required), sender number (pre-filled with their +231 number, editable), amount (pre-filled, locked to the plan price), payment date and time, and a screenshot of the receipt (required).
5. Server validates (below), stores the screenshot in the private payment-evidence bucket and creates the claim as `PENDING_REVIEW`.
6. Member sees “Payment submitted — we’re verifying it” with the expected turnaround **[DECISION]** OD-16.
7. An admin opens the claim, finds the transaction ID and amount in the merchant wallet’s own records, then approves, rejects with a reason, or asks for more information.
8. Approval runs `approve_payment_claim()`: in one transaction the claim becomes `APPROVED`, a payment row is written as `SUCCEEDED`, Casual access is created or extended, an audit entry is written and the member is notified.

Validation at submission (server-side)

- Plan exists, is active and is a mobile money plan.
- Transaction ID matches the expected format for the provider and has never been claimed with status `PENDING_REVIEW`, `NEEDS_INFO` or `APPROVED` (unique index).
- Screenshot: image type checked by magic bytes, size capped, re-encoded, SHA-256 stored. An identical hash on an earlier claim auto-flags the new claim.
- Member has fewer than the maximum pending claims **[DECISION]** OD-21.
- Mobile money purchase is blocked while the member has an active card subscription (BR-41).
- Member account is `ACTIVE` and verified.

Claim states

State	Meaning	Next
PENDING_REVIEW	Waiting for an admin	APPROVED · REJECTED · NEEDS_INFO
NEEDS_INFO	Admin asked the member a question (e.g. clearer screenshot)	PENDING_REVIEW after the member replies
APPROVED	Verified; access granted	Final
REJECTED	Not found, wrong amount, reused or fake; reason stored and shown neutrally	Final; member may submit a new claim
CANCELLED	Withdrawn by the member before review	Final

Why approval needs the wallet record. A screenshot can be edited, reused or borrowed. It is supporting evidence only. An admin approves only after finding the transaction ID and amount in the merchant wallet's own records (wallet app, statement or confirmation SMS).

Admin approval rules

- Only ADMIN or SUPER_ADMIN may decide claims, and never on a claim from their own member account.
- Amount must equal the plan price exactly. Under- or over-payments are handled per **[DECISION]** OD-17.
- Every decision, and every view of a screenshot, writes an audit entry.
- Claims are reviewed oldest first. The dashboard shows the count and the age of the oldest pending claim.

Card subscriptions

- Member picks Weekly or Monthly. Server calls `createCheckout()` and redirects to the processor's hosted page. No card data touches WeKconnectz servers.
- Webhook Edge Function verifies the signature, stores the raw event in `payment_events`, and is idempotent on the processor event ID.
- Successful first charge: subscription ACTIVE, period end set from the processor. Each renewal extends the period.
- Failed renewal: PAYMENT_FAILED. Access continues for a grace period **[DECISION]** OD-20, then EXPIRED.
- Member cancels in the app: cancel at period end; status CANCELLED; access continues until the period ends.
- Reminder notification before each renewal. Receipts and a manage-subscription link in My profile.

Subscription states

State	Applies to	Meaning
PENDING	Card	Checkout started, not confirmed
ACTIVE	Both	Access while <code>expires_at > now()</code>
CANCELLED	Card	Will not renew; access until period end
PAYMENT_FAILED	Card	Renewal failed; grace period running
EXPIRED	Both	Period ended
SUSPENDED	Both	Restricted by an admin or a processor dispute

State	Applies to	Meaning
REFUNDED	Both	Refunded; access ended

Combining sources

- Mobile money pass bought during an active mobile money pass: extends from the current expiry (BR-28).
- Mobile money pass while a card subscription is active: blocked, with the message “You already have an active card subscription.” (BR-41).
- Card subscription started while a mobile money pass is active: **[DECISION]** OD-19.
Recommendation: allowed, starts immediately, and the member is warned that remaining pass time runs at the same time.

Adapter interfaces

```
interface CardProcessor {
  id: string
  createCheckout(input: {
    userId: string; planId: string; successUrl: string; cancelUrl: string
  }): Promise<{ url: string }>
  verifyWebhook(headers: Headers, rawBody: string): VerifiedEvent | null
  cancelAtPeriodEnd(processorSubscriptionId: string): Promise<void>
  getManageUrl?(customerRef: string): Promise<string>
}

// Reserved: replaces manual claims when Orange Money / MTN APIs are available.
interface MobileMoneyProvider {
  id: 'orange_money' | 'mtn_momo'
  initiate(input: {
    paymentId: string; amount: number; currency: string; phone: string
  }): Promise<InitiateResult>
  getStatus(providerRef: string): Promise<ProviderStatus>
  verifyWebhook(headers: Headers, rawBody: string): VerifiedEvent | null
}
```

Immutability

Amount, currency, provider, user and plan on a payment never change after creation. Status changes only through database functions, and every change is appended to payment_events. Refunds are new events, not edits.

Card processor options to research **[DECISION]** OD-4

No processor is confirmed yet. Card networks treat dating as a higher-risk category (merchant category code 7273), so expect underwriting questions. Candidates by type:

Type	Examples to look at	What to confirm
Banks operating in Liberia with e-commerce gateways	Ask your business bank first	Recurring billing with stored cards, hosted checkout, webhooks, dating acceptance
Pan-African payment aggregators	Flutterwave, DPO Pay	Whether they onboard a Liberia-registered merchant, card recurring support, dating acceptance
Processors specialising in dating and adult subscriptions	CCBill, Segpay, Verotel	Entity and location requirements, fees, rolling reserves, weekly billing
Foreign entity with a mainstream processor	Stripe via a US or UK company	Dating acceptance, cost of running the entity, tax

Ask every processor

- Do you onboard a Liberia-registered business? If not, what entity do you need?
- Do you accept dating platforms (MCC 7273)?
- Do you support recurring card billing on a weekly interval, with hosted checkout and signed webhooks?
- Settlement currency and timing, fees, reserves, chargeback handling.
- Do Liberian-issued Visa and Mastercard cards pass your checkout, including 3-D Secure?
- Is a neutral statement descriptor allowed?

Mobile money APIs

Keep the applications with Orange Money and Lonestar Cell MTN moving. Register merchant accounts under a neutral name now; personal wallets are not used. When an API arrives, a MobileMoneyProvider adapter replaces manual claims without changing subscriptions or entitlement logic.

17 Trust & safety

[NEW] This section is what separates a legitimate casual-dating product from a classifieds board for paid sex. It is not optional.

Detection

- Server-side checks on bios, message requests and messages for phone numbers, WhatsApp/social handles, links, currency terms and pricing patterns (e.g. "\$", "USD", "LD", "momo", "per hour", "short time", "transport"). The term list lives in app_settings so moderators can update it.
- Bios and requests: matching text is rejected with a neutral message ("Contact details and prices aren't allowed").
- Messages: matching text is delivered but flagged into a moderation queue. No automatic ban from keyword matches alone.
- Behaviour signals flagged for review: many requests in a short time, the same text sent to many members, frequent short availability windows, many reports.

Payment fraud [NEW]

- A transaction ID or screenshot hash already used on another claim auto-flags the new claim.
- Repeated rejected claims from one member (threshold in app_settings) flag the member for review.
- A fake receipt is grounds for suspension at an admin's discretion.
- Rejection reasons shown to members are neutral; they never reveal which check failed.

Report categories & priority

Category	Priority	Automatic action on report
Appears under 18	HIGH	Removed from discovery and availability immediately, pending review.
Selling or buying sex	HIGH	Flag; auto-hide after threshold.
Asking for money / scam	HIGH	Flag; auto-hide after threshold.
Threats or harassment	HIGH	Flag; auto-hide after threshold.
Fake profile / impersonation	MEDIUM	Flag.

Category	Priority	Automatic action on report
Inappropriate photo	MEDIUM	Photo hidden pending review.
Spam	LOW	Flag.
Other	LOW	Flag.

Auto-hide threshold: distinct reporters within 24 hours, value in app_settings (recommend 3).
Auto-hide removes the user from discovery; it does not ban.

Blocking (kept)

If A blocks B, neither sees the other anywhere, and no messages, likes or requests pass in either direction. Blocks are silent — the blocked user is not told.

Member-facing safety

- Safety page: meet in public places, tell a friend, arrange your own transport, never send money to someone you met here.
- In-chat reminder on first message in each conversation: “Never send money to someone you haven’t met.”
- Report and block reachable from every profile and conversation in two taps.

Legal [VERIFY]

Before launch, get Liberian legal review of the Terms, Privacy Policy, community rules, data retention and the process for responding to law-enforcement requests.

18 Data model

PostgreSQL. All ids UUID. All tables have created_at; mutable ones also updated_at. RLS on every table. Enums as Postgres enums or check constraints.

Table	Key columns	Notes
users	id (= auth.users.id), role, status, suspended_until, last_seen_at, deleted_at	
profiles	user_id PK, display_name, date_of_birth, gender, seeking_genders[], bio, area_id, intent_relationship, intent_casual, is_profile_complete, dob_locked	[CHANGED] no lat/long
areas	id, county, name, active	[NEW] controlled location list
profile_photos	id, user_id, storage_path, sort_order, status, is_primary, rejection_reason, reviewed_by, reviewed_at	
verifications	id, user_id, selfie_storage_path, pose_prompt, status, submitted_at, reviewed_at, reviewed_by, rejection_reason	
interests / user_interests	id, name, slug, active / user_id, interest_id	
user_settings	user_id PK, casual_message_permission (ANYONE NOBODY), notification_prefs	[NEW]
availability	user_id PK, status, start_at, end_at	One row per user
subscription_plans	id, name, source (MOBILE_MONEY CARD), duration_hours, price, currency, renews, processor_price_id?, active	[CHANGED]
subscriptions	id, user_id, plan_id, source, status, starts_at, expires_at (= current period end), auto_renew, cancel_at_period_end, processor_subscription_id?, source_payment_id	[CHANGED]
payments	id, user_id, plan_id, source, provider, provider_transaction_id (unique per provider), claim_id?, amount, currency, status, paid_at	Money fields immutable
payment_events	id, payment_id, type, raw_payload, signature_valid, actor_id?, received_at	Append-only: card webhooks and claim decisions
payment_claims	id, user_id, plan_id, provider, merchant_account_id, reference_code, transaction_id, sender_phone, amount, currency, paid_at, evidence_path, evidence_sha256, status, rejection_reason, member_note, reviewed_by, reviewed_at	[NEW] partial unique (provider, transaction_id) where status in PENDING_REVIEW, NEEDS_INFO, APPROVED
merchant_accounts	id, provider, display_name, number_or_code, active	[NEW] shown in payment instructions
card_customers	user_id, processor, customer_ref	[NEW]
likes	id, sender_id, receiver_id, unique(sender_id, receiver_id)	
passes	id, sender_id, receiver_id, created_at	[NEW] cool-down
matches	id, user_a_id, user_b_id, status, unique(user_a_id, user_b_id), check(user_a_id < user_b_id)	
message_requests	id, sender_id, recipient_id, text, status (PENDING ACCEPTED DECLINED EXPIRED), responded_at	[NEW]

Table	Key columns	Notes
conversations	id, type (RELATIONSHIP CASUAL), match_id?, request_id?, status	
conversation_members	conversation_id, user_id, joined_at, last_read_at	
messages	id, conversation_id, sender_id, body, status, flagged, created_at, read_at	
saved_profiles	user_id, saved_user_id	[NEW]
blocks	id, blocker_id, blocked_id, unique pair	
reports	id, reporter_id, reported_user_id, category, priority, description, status, reviewed_by, reviewed_at	
report_notes	id, report_id, author_id, note	[NEW] internal
moderation_flags	id, entity_type, entity_id, reason, status, reviewed_by	[NEW] auto-detections
notifications	id, user_id, type, payload, read_at	[NEW]
consents	id, user_id, document (TERMS PRIVACY RULES), version, accepted_at	[NEW]
geo_checks	id, phone_hash?, ip_country, phone_country, result, created_at	[NEW]
phone_blocklist	phone_hash, reason, created_by	[NEW] banned numbers
app_settings	key, value (jsonb), updated_by	[NEW] prices, limits, flags, term lists
audit_logs	id, actor_id, action, entity_type, entity_id, metadata, created_at	Append-only

Audit log actions (extended)

USER_SUSPENDED · USER_BANNED · USER_RESTORED · VERIFICATION_APPROVED ·
 VERIFICATION_REJECTED · SELFIE_VIEWED · PHOTO_APPROVED · PHOTO_REJECTED ·
 DOB_CORRECTED · SUBSCRIPTION_MODIFIED · PAYMENT_REFUNDED · PAYMENT_CLAIM_APPROVED ·
 PAYMENT_CLAIM_REJECTED · PAYMENT_CLAIM_NEEDS_INFO · EVIDENCE_VIEWED ·
 REPORT_RESOLVED · SETTING_CHANGED · ROLE_CHANGED · ADMIN_CREATED

19 Business rules

Copy into docs/BUSINESS_RULES.md. Code and tests reference these numbers.

Access

- BR-1 Registration requires request country = LR at signup (\$3).
- BR-2 Registration requires a verified +231 phone number.
- BR-3 One account per verified phone number. Banned numbers cannot register.

Account

- BR-4 User must be 18 or older. Date of birth is locked after entry.
- BR-5 SUSPENDED users cannot use discovery, likes, requests or send messages.
- BR-6 BANNED users cannot authenticate.
- BR-7 DELETED users never appear anywhere and cannot be rediscovered.

Photos

- BR-8 Minimum 3 APPROVED photos to appear in any discovery.
- BR-9 Only APPROVED photos count toward the minimum.
- BR-10 Verification selfie is private and never appears in discovery.
- BR-11 Photos are served only via signed URLs after a server-side access check.
- BR-12 All uploads have EXIF metadata stripped.

Verification

- BR-13 Every member must be VERIFIED to appear in Relationship or Casual discovery.
- BR-14 Verification status is shown as a badge.
- BR-15 REJECTED verification prevents discovery and availability.

Casual

- BR-16 Casual discovery requires an active pass.
- BR-17 A user appears in the pool only if AVAILABLE, inside their window, and holding an active pass.
- BR-18 Availability windows end automatically; users can leave the pool at any time.
- BR-19 Availability is never visible outside the pool.
- BR-20 Exact location is never collected or exposed.

Messaging

- BR-21 Casual contact starts with a message request from a verified pass-holder to an AVAILABLE member.
- BR-22 Recipient can accept, decline or block.
- BR-23 Relationship messaging requires a mutual match.
- BR-24 Blocked users cannot see or contact each other in any way.
- BR-25 Without an active pass, CASUAL conversations are read-only.

Subscription

- BR-26 Casual access begins only after confirmed payment: an approved claim (mobile money) or a verified processor event (card).
- BR-27 Access ends at expires_at.
- BR-28 A mobile money pass bought during an active mobile money pass extends from the current expiry.
- BR-29 Payment money fields are immutable; status changes are logged as events.
- BR-30 Entitlement is derived from server-side state, never trusted from the client.

Safety

- BR-31 Bios and requests containing contact details or prices are rejected.
- BR-32 An under-18 report removes the user from discovery immediately pending review.
- BR-33 Reaching the report threshold auto-hides a user pending review.
- BR-34 Every sensitive staff action is audit-logged.

Payment claims and cards

- BR-35 A transaction ID can be approved only once per provider.
- BR-36 A claim can be approved only if the amount equals the plan price exactly.
- BR-37 Only ADMIN or SUPER_ADMIN may decide a claim, never on their own member account.
- BR-38 An admin approves only after finding the transaction in the merchant wallet records; a screenshot alone is never enough.
- BR-39 A member may have only a limited number of claims pending at once (OD-21).
- BR-40 A cancelled card subscription keeps access until the period ends; a failed renewal enters PAYMENT_FAILED, then EXPIRED after the grace period.
- BR-41 A mobile money pass cannot be bought while a card subscription is active.

20 Screen map

PUBLIC Landing About Pricing Safety Terms Privacy Community rules Login Register Region blocked	ONBOARDING Age gate Phone + OTP Rules & terms Basic profile Intent Interests Bio Photos Verification selfie Under review	APP Home Relationship Discover Likes received Matches Casual Available Now (+ filters) Saved Get access Choose plan (mobile money card) Mobile money instructions [NEW] Submit payment claim [NEW] Claim status [NEW] Card checkout (processor) [NEW] Messages Conversations · Requests · Conversation Notifications Member profile (view) My profile Edit profile · Photos Availability · Verification Subscription & payments [CHANGED] (claims, receipts, cancel card plan) Privacy & messaging · Blocked users Settings · Delete account
ADMIN (staff accounts, MFA) Dashboard Users Verification queue Photo queue Flags queue Payment claims queue [NEW] Subscriptions Payments & events Reports Analytics Audit logs Settings (plans, merchant accounts, limits, flags) Staff		

21 Admin console

Screen	Contents & actions
Dashboard	Users, verified users, active Casual access by source and plan, users available now, revenue by source, open reports, queue sizes, pending claims and age of the oldest.
Users	Search by name, ID, phone (hashed lookup), status, verification, access, availability. View, suspend (with end date), ban, restore, view reports and audit trail.
Verification queue	Profile photos beside the selfie and pose prompt. Approve / reject with reason. Oldest first.
Photo queue	Pending photos. Approve / reject with reason.
Flags queue	Auto-detected bios, requests, messages, behaviour signals and reused payment evidence.
Payment claims [NEW]	Oldest first. Claim details beside the screenshot. Checklist: transaction found in merchant wallet · amount matches plan · date plausible · sender number noted. Approve · Reject (reason) · Needs info (message to member). Earlier claims from the same member shown.
Reports	Priority-sorted. Investigate, internal notes, suspend, ban, dismiss, resolve.
Subscriptions	Active, cancelled, payment failed, expired, suspended, refunded. Manual extension with reason (audited).
Payments & events	Every payment with source, provider, amount, status and its event history; card webhook log.

Screen	Contents & actions
Settings	Plans and prices, merchant accounts, limits and thresholds, detection terms, interests, areas, report categories, feature flags.
Audit logs	Filter by actor, action, entity, date. Read-only.

22 Security architecture

Area	Requirement
Authorization	RLS on every table + server-side entitlement checks. pgTAP tests prove a user cannot read another user's private rows, selfies, or pool data without a pass.
Secrets	Service-role key and provider secrets only in server-only modules and Edge Function env. Never in NEXT_PUBLIC_*.
Storage	Private buckets only. Signed URLs with short TTL. Storage paths never sent to clients.
Uploads	Magic-byte validation, size limits, re-encode, EXIF strip, per-user rate limit.
Webhooks	Card processor webhooks: signature verification, raw-body logging, idempotent on processor event ID, replay protection.
Payment evidence	Private bucket; signed URLs for reviewers only; every view audited; SHA-256 stored to catch reused images; retention per OD-18.
Rate limits	OTP sends per phone and per IP; requests, likes, reports, uploads per user per day.
Staff	Separate accounts, MFA required, server-side role checks on every admin route and action, all sensitive actions audited.
Privacy	No precise location. PII excluded from logs and analytics. Data export and deletion on request.
Headers	CSP, HSTS, X-Frame-Options DENY, Referrer-Policy strict-origin-when-cross-origin.
Backups	Supabase point-in-time recovery or daily backups; restore tested before launch.

23 Build plan

One sprint at a time. Each ends with a working, tested feature and owner sign-off. Card work does not block launch.

#	Sprint	Acceptance criteria (summary)
0	Foundation & agent docs	Repo, CLAUDE.md, docs/, Supabase project, migrations, CI (lint, type-check, tests), Sentry.
1	Auth, geo & age gate	+231 OTP; non-LR signup blocked before OTP; geo_checks recorded; under-18 blocked; users, roles and states with RLS.
2	Profile & onboarding	Steps 4–8 with resume; areas, interests, consents stored; bio detection rejects contacts and prices.
3	Photos & private storage	Upload pipeline with EXIF strip; photo states; photo moderation screen; signed-URL access tests.
4	Verification	Pose-prompt selfie; verification queue; ACTIVE when verified with 3 approved photos; selfie views audited.
5	Safety core	Blocking, reporting, auto-hide threshold, under-18 instant hide, flags queue, phone blocklist.
6	Relationship + messaging core	Discover, like/pass, atomic matching, chat over Realtime, text only, detection flags.
7	Mobile money access	Plans from DB; merchant accounts; claim submission with evidence; claims queue; approve_payment_claim(); stacking; expiry; notifications.
7b	Card subscriptions	After the processor is chosen: adapter, hosted checkout, webhooks, renewals, failures, cancellation. Can ship after launch.
8	Availability	Now / scheduled windows with caps; eligibility; query-time pool membership; cron cleanup.
9	Casual discovery & requests	Available Now with filters; requests with caps and cool-down; read-only on expiry; saved profiles.
10	Admin console	All §21 screens; staff MFA; settings; audit log viewer.
11	Notifications	Requests, matches, messages, review outcomes, claim decisions, expiry and renewal reminders.
12	Hardening & launch	Full RLS test pass, security headers, rate limits, backup restore drill, legal review sign-off.

24 Out of scope for the MVP

[NEW] An agent must not build these, even if they seem helpful.

Item	Status	Reason
WhatsApp channel of any kind (bot, receipts, notifications)	Do not build	Meta's WhatsApp Business policy prohibits dating services; violations can bar the whole organisation from WhatsApp business products.
Coins or in-app wallet	Removed	Owner decision.
Voucher codes	Removed	Access activates directly after approval.
AI or automatic approval of mobile money claims	Do not build	Owner decision: admins approve manually. A screenshot cannot prove payment.
In-app AI assistant for payments	Later, optional	MVP uses a guided form. AI may later pre-fill fields from a screenshot; it never approves.
Recurring mobile money	Not possible	No recurring debit without provider APIs.
Direct Orange Money / MTN API integration	Later	Waiting on providers; MobileMoneyProvider interface reserved.
Images, voice or links in chat	Not in MVP	Moderation load and safety.
Precise location / GPS	Never	Privacy and safety.
Native Android / iOS apps	Phase 2 / 3	Web first.
Diaspora access	Not in MVP	Owner decision.

25 Open decisions

An agent must not fill these in. Resolve each in docs/DECISIONS.md before the sprint that needs it.

ID	Decision	Recommendation	Needed by
OD-1	Price of every plan (mobile money and card)	—	Sprint 7
OD-2	Currency: USD, LRD or both	Match what the merchant wallets collect	Sprint 7
OD-3	Photo visibility in free Relationship mode	Option A (\$11)	Sprint 3
OD-4	Card processor	Research per \$16	Sprint 7b
OD-5	ID document check before Casual eligibility	Yes	Sprint 4
OD-6	Verification selfie retention	Delete 90 days after approval	Sprint 4
OD-7	Data retention after account deletion	Grace period, then purge; keep payment and audit records	Sprint 10
OD-8	Availability window max length / lead time	12 hours / 7 days	Sprint 8
OD-9	Message request daily cap and decline cool-down	—	Sprint 9
OD-10	Daily like cap for Relationship	—	Sprint 6
OD-11	Hosting: Vercel or own VPS	Vercel for MVP speed	Sprint 0
OD-12	Store raw signup IP or country only	Country only	Sprint 1

ID	Decision	Recommendation	Needed by
OD-13	Refund policy	—	Sprint 7
OD-14	Roaming Liberians blocked at signup — intended?	—	Sprint 1
OD-15	Offer Monthly on card as well as Weekly	Yes	Sprint 7b
OD-16	Claim review turnaround shown to members; staff review hours	Publish fixed review hours	Sprint 7
OD-17	Under- or over-payment handling	Reject with reason; refund manually	Sprint 7
OD-18	Payment screenshot retention	Delete 90 days after decision; keep claim record	Sprint 7
OD-19	Card subscription during an active mobile money pass	Allow; runs concurrently; warn member	Sprint 7b
OD-20	Grace period after failed card renewal	Short, e.g. 48 hours	Sprint 7b
OD-21	Maximum pending claims per member	2	Sprint 7
OD-22	Can members add a reference note to Orange / MTN payments?	Check with providers [VERIFY]	Sprint 7